ZIMSEC MS 6042/1

MR SHARE ONLINE

REVISION .ft

[TUKS. t: 1~2 testing]

General Certificate of Education Advanced Level

PURE MATHEMATICS **6042/1**

PAPER 1 SOLUTION

JUNE 2024 SESSION 3 hours

Additional materials:
- Answer paper
- Graph paper
- List of formulae MF7
- Electronic calculator (Non-programmable)

Time 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

INFORMATION TO CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part of question.

The total number of marks for this paper is 120.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given to 2 significant figures.

Answer ***all*** questions.

The use of a scientific calculator is expected, where appropriate.

This question paper consists of 7 printed pages and 1 blank page.

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1. A sequence is defined by $U_{n+1} = 3U_n + 1$
   (i) Write down the first tour terms of the sequence, given that $U_1 = 0$. [2]
   (ii) Hence describe the behaviour of the sequence. [1]
   (iii) Express U_3 in terms of U_1. [2]

SOLUTION

(i) $U_{n+1} = 3U_n + 1$

$U_1 = 0$

$U_2 = 3U_1 + 1$

$= 3(0) + 1$

$= \mathbf{1}$

$U_3 = 3(U_2) + 1$

$= 3(1) + 1$

$= \mathbf{4}$

$U_4 = 3(U_3) + 1$

$= 3(4) + 1$

$= \mathbf{13}$

(ii) The sequence is diverging, i.e $[0, +\infty)$

(iii) $U_3 = 3(U_2) + 1$

$= 3(3U_1 + 1) + 1$

$= 9U_1 + 3 + 1$

$= \mathbf{9U_1 + 4}$

2. The variable x and y are connected by the equation $y = ae^{bx}$, where a and b are constants. When the values of Iny are plotted against values of x, a straight is obtained passing through the points $(0; In2)$ and $(3; In5)$.
   Calculate the values of a and b in exact form. [6]

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SOLUTION

$$y = ae^{bx}$$
$$Iny = In(ae^{bx})$$
$$Iny = Ina + Ine^{bx}$$
$$\boldsymbol{Iny = bx + Ina}$$

In relation the line; $y = mx + c$

$y = Iny;\ m = b;\ x = x;\ c = Ina$

Gradient $b = m = \frac{In2 - In5}{0-3} = -\frac{1}{3}\boldsymbol{In}\left(\frac{2}{5}\right)$

$$Ina = c = In2$$
$$e^{Ina} = e^{In2}$$
$$\boldsymbol{a = 2}$$

3. Express $\frac{2x-3}{x^2(x^2-4)}$ in partial fractions. [7]

SOLUTION

$$\frac{2x-3}{x^2(x^2-4)} = \frac{2x-3}{x^2(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-2)} + \frac{D}{(x+2)}$$
$$2x - 3 = Ax(x^2-4) + B(x^2-4) + Cx^2(x+2) + Dx^2(x-2)$$

When $x = 0$;

$$-3 = B(-4)$$
$$\boldsymbol{B = \frac{3}{4}}$$

When $x = 2$;

$$2(2) - 3 = C(2^2)(2+2)$$
$$1 = C(16)$$
$$\boldsymbol{C = \frac{1}{16}}$$

When $x = -2$;

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$$2(-2) - 3 = D(-2)^2(-2-2)$$

$$-7 = -D16$$

$$\boldsymbol{D = \frac{7}{16}}$$

When $x = 1$;

$$2(1) - 3 = A(-3) + B(-3) + C(3) + D(-1)$$

$$-1 = -3A - 3\left(\frac{3}{4}\right) + 3\left(\frac{1}{16}\right) - \frac{7}{16}$$

$$\boldsymbol{A = -\frac{1}{2}}$$

$$\boldsymbol{\therefore \frac{2x-3}{x^2(x^2-4)} = -\frac{1}{2x} + \frac{3}{4x^2} + \frac{1}{16(x-2)} + \frac{7}{16(x+2)}}$$

4. Solve the equation $\frac{e^{2x}+e^{-2x}}{5} = 1$, giving the answer in exact form. [7]

SOLUTION

$$\frac{e^{2x}+e^{-2x}}{5} = 1$$

$$e^{2x} + e^{-2x} = 5$$

Let, $e^{2x} = a$

$$a^2 - 5a + 1 = 0$$

$$a = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(1)}}{2}$$

$$= \frac{5 \pm \sqrt{21}}{2}$$

$$a = \frac{5-\sqrt{21}}{2} \quad \text{or} \quad a = \frac{5+\sqrt{21}}{2}$$

But, $a = e^{2x}$, hence

$$e^{2x} = \frac{5-\sqrt{21}}{2} \quad \text{or} \quad e^{2x} = \frac{5+\sqrt{21}}{2}$$

$$\boldsymbol{x = \frac{1}{2}In\left(\frac{5-\sqrt{21}}{2}\right) \text{ or } x = \frac{1}{2}In\left(\frac{5+\sqrt{21}}{2}\right)}$$

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5. Find the set of values of x for which $(x-3)^2 > 2x+1$, giving answers in exact form. [7]

SOLUTION

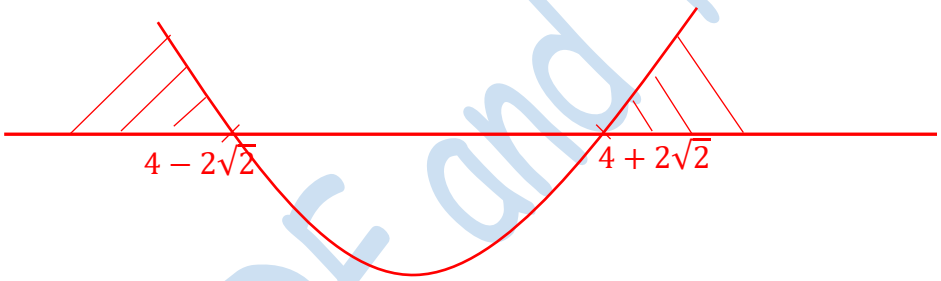
$$x^2 - 6x + 9 - 2x - 1 > 0$$

$$x^2 - 8x + 8 > 0$$

Critical values of x;

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(1)(8)}}{2}$$

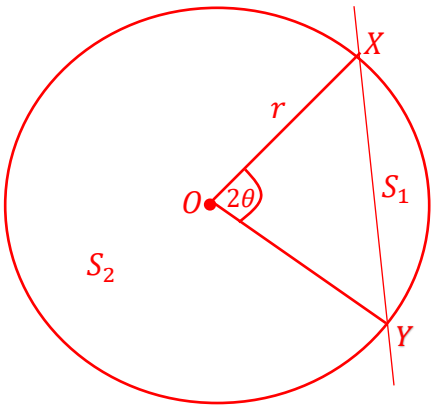
$$x = \frac{8 \pm \sqrt{32}}{2} = 4 \pm 2\sqrt{2}$$



Hence, $x < 4 - 2\sqrt{2} \cup x > 4 + 2\sqrt{2}$

6. Chord XY subtends an angle of 2θ radians at the centre of a circle radius r and centre O. Given that the area of the minor segment is $\frac{1}{6}$ of the area of the major segment, show that $sin2\theta = \frac{2}{7}(7\theta - \pi)$. [7]

SOLUTION



SOLUTION

$$S_1 = \frac{1}{6}S_2$$

$$\frac{1}{2}r^2(2\theta - \sin 2\theta) = \frac{1}{6}.\frac{1}{2}r^2[(2\pi - 2\theta) - \sin(2\pi - 2\theta)]$$

$$2\theta - \sin 2\theta = \frac{1}{6}[2\pi - 2\theta - (\sin 2\pi \cos 2\theta - \sin 2\theta \cos 2\pi)]$$

$$2\theta - \sin 2\theta = \frac{1}{6}[2\pi - 2\theta + \sin 2\theta]$$

$$12\theta - 6\sin 2\theta = 2\pi - 2\theta + \sin 2\theta$$

$$7\sin 2\theta = 14\theta - 2\pi$$

$$\boldsymbol{\sin 2\theta = \frac{2}{7}(7\theta - \pi)} \ \ \textbf{[shown]}$$

7.

(a) Expand $(1-2x)^{\frac{1}{2}}$ up to and including the term in x^2, simplifying coefficients. [3]

(b) State the values of x for which the expansion is valid. [1]

(c) By putting $x = \frac{1}{9}$ in the expansion in part (a), show that $\sqrt{7} \approx \frac{143}{54}$. [4]

SOLUTION

(i) $(1-2x)^{\frac{1}{2}} = 1 + \frac{1}{2}(-2x) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}(-2x)^2 + \cdots$

$$= \boldsymbol{1 - x - \frac{1}{2}x^2 + \cdots}$$

(ii) For validity;

$$|-2x| < 1$$

$$\boldsymbol{-\frac{1}{2} < x < \frac{1}{2}}$$

(iii) $\left(1 - 2\left(\frac{1}{9}\right)\right)^{\frac{1}{2}} \approx 1 - \frac{1}{9} - \frac{1}{2}\left(\frac{1}{9}\right)^2$

$$\left(\frac{7}{9}\right)^{\frac{1}{2}} \approx \frac{143}{162}$$

$$\boldsymbol{\sqrt{7} \approx 3\left(\frac{143}{162}\right) = \frac{143}{54}} \ \ \textbf{[shown]}$$

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8. The equation of line l is $y + 3x - 9 = 0$.

Find the,

(a) equation of the line m, that is perpendicular to line l and passes through the point A(1:6) in the form $ax + by + c = 0$. [4]

(b) equation of the line parallel to l passing through the point B(5:6). [2]

(c) perpendicular distance of the point B(5;6) from the line m. [3]

SOLUTION

(a) $y = -3x + 9, \quad \therefore m = -3$

$$mn = -1 \quad \text{[product of gradient of perpendicular lines]}$$

$$n = \frac{-1}{m} = -\frac{1}{(-3)} = \frac{1}{3}$$

$$y - y_1 = n(x - x_1) \text{ [equation of the normal to } l\text{]}$$

$$y - 6 = \frac{1}{3}(x - 1)$$

$$y = \frac{1}{3}x - \frac{17}{3}$$

$$\boldsymbol{x - 3y + 17 = 0}$$

(b) $y - y_1 = m(x - x_1)$ [parallel lines have same gradient]

$$y - 6 = -3(x - 5)$$

$$\boldsymbol{y = -3x + 21}$$

(c) $x - 3y + 17 = 0 \ldots.(1)$

$y = -3x + 21 \ldots.(2)$

Solving equation (1) and (2) yields;

$$x = 4.6; \quad y = 7.2$$

Now distance from B(5; 6);

$$d = \sqrt{(5 - 4.6)^2 + (6 - 7.2)^2}$$

$$= \frac{2}{5}\sqrt{10}$$

$$= \boldsymbol{1.3\ units\ [2s.f]}$$

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9.

(a) Express $x^2 + 6x + 5$ in the form $(ax + b)^2 + c$, where a, b and c are constant. [2]

(b) Hence or otherwise state the turning point of the graph of $y = x^2 + 6x + 5$. [1]

(c) Find the set of values taken by $y = \frac{1}{x^2+6x+5}$ for real values of x. [7]

SOLUTION

(a) $x^2 + 6x + 5 = x^2 + 6x + \left(\frac{6}{2}\right)^2 + 5 - \left(\frac{6}{2}\right)^2$

$= \boldsymbol{(x + 3)^2 - 4}$

$\boldsymbol{a = 1, b = 3 \text{ and } c = -4}$

(b) The turning point of the graph is; $\boldsymbol{(-3, -4)}$

(c) $y = \frac{1}{x^2+6x+5} = \frac{1}{(x+1)(x+5)}$

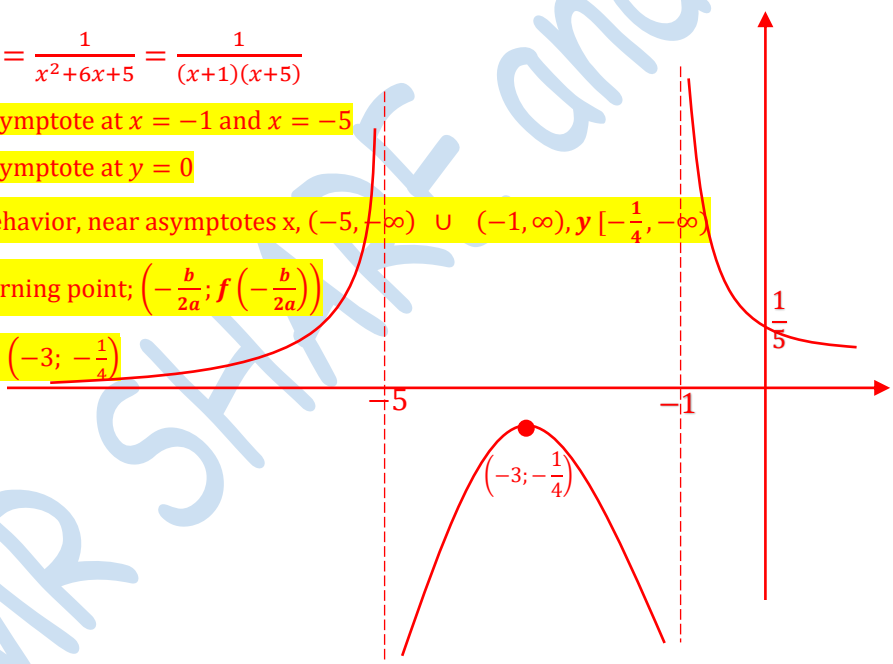
asymptote at $x = -1$ and $x = -5$

asymptote at $y = 0$

behavior, near asymptotes x, $(-5, -\infty) \cup (-1, \infty)$, $\boldsymbol{y}\ [-\frac{1}{4}, -\infty)$

turning point; $\left(-\frac{b}{2a}; f\left(-\frac{b}{2a}\right)\right)$

i.e $\left(-3; -\frac{1}{4}\right)$



Since $x^2 + 6x + 5 = (x + 1)(x + 5)$, is always positive except at $x = -1$ and $x = -5$.

Then, $y = \frac{1}{x^2+6x+5} > 0 \cup y \leq -\frac{1}{4}$ for real values of x, i.e $\boldsymbol{y > 0 \cup y \leq -\frac{1}{4}}$

10. Given that $f(x) = x^4 + x^3 + 23x^2 + 25x - 50$,

(a) show that $-5i$ is a root of $f(x) = 0$, [3]

(b) hence solve the equation $f(x) = 0$. [7]

SOLUTION

(a) $f(x) = x^4 + x^3 + 23x^2 + 25x - 50$

$$f(-5i) = (-5i)^4 + (-5i)^3 + 23(-5i)^2 + 25(-5i) - 50$$
$$= 625 + 125i - 575 - 125i - 50$$
$$= 0$$

(b) $(x - 5i)(x + 5i) = x^2 - (-5i)^2$ [note the other root is the conjugate of $-5i$]

$= x^2 + 25$ [factor]

$$\begin{array}{r|l} & x^2 + x - 2 \\ \hline (x^2 + 25) & x^4 + x^3 + 23x^2 + 25x - 50 \\ & -(x^4 + 25x^2) \\ \hline & x^3 - 25x^2 + 23x^2 \\ & x^3 - 2x^2 + 25x - 50 \\ & -(x^3 \qquad + 25x) \\ \hline & -2x^2 - 50 \\ & -(-2x^2 - 50) \\ \hline & 0 \end{array}$$

$x^2 + x - 2 = 0$

$(x + 2)(x - 1) = 0$

$x = -2 \ or \ 1$

Hence the solutions for x are; $-5i$; $5i$; $-2, 1$

11. The functions f, g *and* h defined as:

$f: x \to x^2 + 2x \qquad x \in \mathbb{R},\ x \geq -1$

$g: x \to 4x + 1 \qquad x \in \mathbb{R},\ -2 \leq x \leq 3$

$h: x \to \frac{1}{x-3} \qquad x \in \mathbb{R},\ x \neq 3$

(a) Find

(i) $gh(x)$, stating clearly its domain, [3]

(ii) $f^{-1}(x)$. [4]

(b)

(i) Sketch the graph of $y = |g(x)|$, showing clearly the intercept and end points. [2]

(ii) Hence state the range of $|g(x)|$. [1]

SOLUTION

(a)

(i) $gh(x) = 4\left(\frac{1}{x-3}\right) + 1$

$= \mathbf{1} + \frac{\mathbf{4}}{\boldsymbol{x}-\mathbf{3}},\ \boldsymbol{x} \in \mathbb{R}, \boldsymbol{x} \neq \mathbf{3}$ [domain of composite function is the same as the domain of the input function not the carrying function.]

(ii) let $y = f(x)$

$y = x^2 + 2x + 1^2 - 1^2$

$y = (x+1)^2 - 1$

$x = -1 \pm \sqrt{y+1}$

$\therefore \boldsymbol{f^{-1}(x) = -1 + \sqrt{x+1}},\quad \boldsymbol{x} \in \mathbb{R}, \boldsymbol{x} \geq -\mathbf{1}$

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NOTES

When finding the inverse of the function $f: x \to x^2 + 2x \quad x \in \mathbb{R},\ x \geq -1$ we need to choose the branch of the square root that maintains the domain restriction. Here's why we chose the positive branch;

The original function $f: x \to x^2 + 2x \quad x \in \mathbb{R},\ x \geq -1$. We solve for x from the equation $(y = x^2 + 2x)$, resulting in;

$$x = -1 \pm \sqrt{y+1}$$

We have two potential solutions;

$$x = -1 + \sqrt{y+1} \quad \text{or} \quad x = -1 - \sqrt{y+1}$$

Since $x \geq -1$;

- the solution $x = -1 - \sqrt{y+1}$ will always yield values less than or equal to (-1)because $-\sqrt{y+1} \leq 0$. This branch does not satisfy $x \geq -1$, except at the boundary point where $y = -1$, which gives $x = -1$
- the solution $x = -1 + \sqrt{y+1}$ yields values greater than or equal to (-1) because $\sqrt{y+1} \geq 0$. This branch satisfies $(x \geq -1)$

Therefore the correct inverse function respecting the domain restriction is;

$$\boldsymbol{f^{-1}(x) = -1 + \sqrt{x+1}}$$

This ensures the inverse function maps to the appropriate range of values for $(x \geq -1)$.

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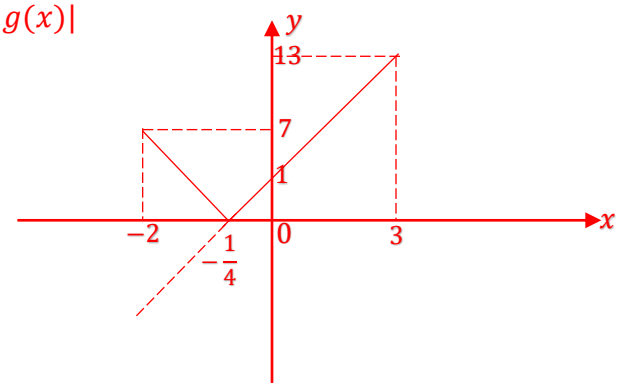
(b)

(i) $y = |g(x)|$



(ii) Range of $|g(x)|$; $\mathbf{0 \leq g(x) \leq 13}$

12.

(a) Prove the identity

$$\frac{\cos\theta}{1-\cos\theta} - \frac{\cos\theta}{1+\cos\theta} = 2\cot^2\theta$$

[4]

(b) Hence or otherwise solve the equation $\frac{\cos\theta}{1-\cos\theta} - \frac{\cos\theta}{1+\cos\theta} = 1$ for $\theta° \leq \theta \leq 360°$ giving your answer to the nearest degree. [7]

SOLUTION

(a) $\frac{\cos\theta}{1-\cos\theta} - \frac{\cos\theta}{1+\cos\theta} = 2\cot^2\theta$

RHS;

$$= \frac{\cos\theta(1+\cos\theta) - \cos\theta(1-\cos\theta)}{1-\cos^2\theta}$$

$$= \frac{\cos\theta + \cos^2\theta - \cos\theta + \cos^2\theta}{\sin^2\theta}$$

$$= 2\left(\frac{\cos^2\theta}{\sin^2\theta}\right)$$

$$= \mathbf{2\cot^2\theta \equiv RHS \ (shown)}$$

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(b) $\frac{\cos\theta}{1-\cos\theta} - \frac{\cos\theta}{1+\cos\theta} = 1$ for $\theta° \leq \theta \leq 360°$

$$2\cot^2\theta = 1$$

$$\frac{1}{\tan^2\theta} = \frac{1}{2}$$

$$\tan^2\theta = 2$$

$$\tan\theta = \pm\sqrt{2}$$

$$\tan\theta = -\sqrt{2} \quad \text{or} \quad \tan\theta = \sqrt{2}$$

$$\theta = \tan^{-1}(\pm\sqrt{2}) \quad \text{for} \quad \theta° \leq \theta \leq 360°$$

$$\theta = \pm 54.73561032° \ (PV)$$

$$\boldsymbol{\theta = \pm 54.73561032° + 180°(n)}$$

When $n = 0$,

$$\theta = 54.73561032°$$

When $n = 1$

$$\theta = 125.2643897° \text{ or } 234.7356103°$$

When $n = 2$

$$\theta = 305.2643897°$$

$\therefore \boldsymbol{\theta = 55°; 125°; 235°; 305°}$ **[to the nearest degree]**

13.

(c)

(i) By sketching two appropriate graphs, show that $x^3 + x - 4 = 0$ has one real root. [4]

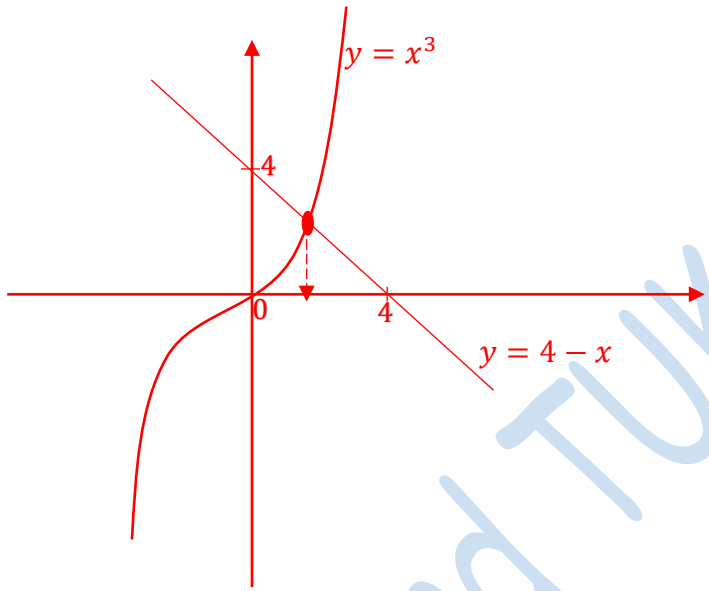
(ii) Show that there is a root between 1 and 1.5. [3]

(d) By using the formula $x = \sqrt{\frac{4}{x_n} - 1}$ and taking x_1 as 1.37, approximate the root of $x^3 + x - 4 = 0$ correct to 2 decimal places. [4]

SOLUTION

(a)

(i) Sketch the graph of $y = x^3$ and $y = 4 - x$ on the same plot;



The graph has one real root, since there is one point of intersection.

(ii) Let $f(x) = x^3 + x - 4$

$$f(1) = (1)^3 + 1 - 4 = -2 < 0$$

$$f(1.5) = (1.5)^3 + 1.5 - 4 = 0.875 > 0$$

Since there is a change of sign i.e $f(1) < 0$ and $f(1.5) > 0$, there lies a root between [1, 1.5].

(b) $x = \sqrt{\frac{4}{x_n} - 1}$, $x_1 = 1.37$

$$x_2 = \sqrt{\frac{4}{x_1} - 1} = \sqrt{\frac{4}{1.37} - 1} = 1.385535286$$

$$x_3 = 1.37367058$$

$$x_4 = 1.382716995$$

$$x_5 = 1.37581073$$

$$x_6 = 1.381078078$$

$\therefore$ **The root is; 1.38 [to 2 decimal places]**

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14. The equation of a curve is $y^2 - 3xy + 2x^2 = 6$

(a) Show that $\frac{dy}{dx} = \frac{3y-4x}{2y-3x}$. [4]

(b) Find the equation of the tangent to the curve at $x = 1$ and $y > 0$. [8]

SOLUTION

(a) $y^2 - 3xy + 2x^2 = 6$

$$2y\frac{dy}{dx} - 3y - 3x\frac{dy}{dx} + 4x = 0$$

$$\frac{dy}{dx}(2y - 3x) = 3y - 4x$$

$$\boldsymbol{\frac{dy}{dx} = \frac{3y - 4x}{2y - 3x}} \text{ [shown]}$$

(b) Gradient is at $m = \frac{dy}{dx}$, when $x = 1$ and $y > 0$

When $x = 1$;

$$y^2 - 3(1)y + 2(1)^2 = 6$$

$$y^2 - 3y - 4 = 0$$

$$(y - 4)(y + 1) = 0$$

$$y = 4 \text{ or } -1$$

Since $y > 0$, $\therefore y = 4$, and the point is, $(1; 4)$

$$\frac{dy}{dx} = \frac{3y - 4x}{2y - 3x}$$

$$m = \frac{dy}{dx} = \frac{3(4) - 4(1)}{2(4) - 3(1)} = \frac{8}{5}$$

The equation of the tangent to the curve at $(1; 4)$ is,

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \frac{8}{5}(x - 1)$$

$$y = \frac{8}{5}x + \frac{12}{5}$$

$$\boldsymbol{8x - 5y + 12 = 0 \text{ or } [5y - 8x = 12]}$$

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FEEL FREE TO CONTACT ME FOR ANY ADJUSTMENTS, CLARIFICATIONS AND ASSISTANCE!

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"Concept before anything!", Author

Proverbs 11 vs. 2

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NB: If you click any of the links below and it fails to direct you straight to the respective group, then first copy the whole link of any group and, place it in any WhatsApp/Telegram chat/group, then click again and join!

WhatsApp group 1 link: https://chat.whatsapp.com/GkRjwDjJr808QbpRRZ3wkf

WhatsApp group 2 link: https://chat.whatsapp.com/Hv3KgLep5etBUdE1RjPkDF

WhatsApp group 3 link: https://chat.whatsapp.com/BTOJO6v6yi5HGWH2LfCg7t

WhatsApp group 4 link: https://chat.whatsapp.com/IzJ7csROuVhANBmHkCifAT

WhatsApp group 5 link: https://chat.whatsapp.com/DJeLdwJWCmi6J8PeNPoLOk

Telegram group link: t.me/shareALmathematics

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